

Engineering Assignment: Airline Service Assistant

Accompanying material: policy documents for 3 fictional airlines, supplied as separate PDFs alongside this brief.

Overview

Build a conversational assistant that helps travelers understand airline policies and manage their bookings.

We provide policy documents from several fictional airlines. Your assistant should use these documents and interact with a backend service to support relevant booking operations.

Treat this as a product being prepared for an initial trial with real users. You decide which scenarios matter most and how to address them.

Scope

The assistant should support:

- Answering questions about airline policies.
- Retrieving booking information.
- Helping users change or cancel a booking and request a refund, where applicable.

Your assistant covers every airline in the pack, not one of them. Travellers do not say up front which airline they flew, and the airlines do not have the same rules.

You may implement the backend APIs as mocks. Define your own API contracts and sample data. Mock operations should maintain enough state to demonstrate complete interactions rather than always returning a fixed success response.

The assistant may be used by more than one person at the same time. Accounting for that is a bonus rather than a requirement; a simple implementation that states its assumptions is perfectly acceptable.

A CLI, a simple web interface, or a chat API is sufficient. Visual polish is not a priority.

Technical Choices

Use any language, framework, model, or development tools you prefer. AI coding tools are welcome.

Beyond the scope above, you decide what is necessary to make the assistant useful and dependable. State any assumptions you make.

Deliverables

- 1 A runnable project with setup instructions and sample data.
- 2 A way for us to reproduce what you saw working: screenshots of real conversations, or written instructions precise enough to follow without guessing, naming which cases to run, what to do in the interface, and what the assistant should do in response.
- 3 Lightweight evaluation and observability:
 - 8 to 10 test cases in the repository, each stating the behaviour you expected and whether your implementation produced it. A JSON or CSV file, or a short Markdown table, is enough. Choose the cases yourself; which behaviours you judged worth testing is part of what we read. Include paths that fail as well as paths that work.
 - A concise execution trace for one conversation: which sources were retrieved, which actions

were requested and what they returned, and any handoff decision. Do not expose hidden reasoning.

4 A short explanation covering:

- The scenarios you prioritized and why.
- Your main design decisions and trade-offs.
- How you assessed whether the assistant works well.
- Known limitations and what you would address next.

Please spend no more than 6 hours. A focused implementation with clear reasoning is preferable to a large, unfinished system; if you get there sooner, stop and tell us what you deliberately left out. Briefly note your actual time spent.

Review

We will assess problem identification, prioritization, implementation quality, and the evidence supporting your decisions.

During the follow-up interview, we will discuss your implementation and explore how it behaves under additional scenarios.